Project Report
on

MESA Stellar Modelling for Asteroseismology

submitted by

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Nurturing talent...promoting excellence

मौलिक विज्ञान प्रकर्ष केन्द्र

Certificate

This is to certify that Vishal Sai Vetrivel, student of UM DAE CEBS, has undertaken project work from January 2026 to May 2026 under the guidance of Prof. Shravan Hanasoge, TIFR.

This submitted report titled MESA Stellar Modeling for Asteroseismology is towards the academic requirements of the Integrated M. Sc. course at UM DAE CEBS.

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Abstract

The determination of stellar age for far away stars is a challenge due to the scant amount of data available to us. Numerical modelling of stars is one of the best methods to make use of the limited data we have. Improving our methods for simulation and by bettering our understanding of its drawbacks allows us to make better use of the data we have. In this project we discuss two models that can be improvements over the usual mixing length theory used for stellar convection and methods of surface correction.

Contents

1. Introduction

Asteroseismology is the study of oscillations in stars. The path of sound inside a star depends on the local speed of sound, which in turn depends on the local temperature and chemical composition. These oscillations are sensitive to different parts of the star which allow the study of the structure of the star which would normally be impossible to infer from overall properties such as brightness or surface temperature.

Asteroseismology is similar to Helioseismology as the underlying physics is the exact same. The difference between the fields arises from the difference in data available to study. In the case of the Sun, we are relatively close as observers and as a result are able to obtain spatially resolved data. This spatial resolution allows us to extract more information from the oscillations themselves. In the case of faraway stars, we only have access to surface integrated variables such as luminosity. This severely limits the information that can be extracted.

1.1 Data used

The data for this thesis were taken from the Kepler space telescope (Borucki et al., 2010). The Kepler Space Telescope was launched by NASA in 2009. It has since been retired in 2018, due to the fuel for its reaction control system running out. It was succeeded by the Transiting Exoplanet Survey Satellite or TESS, launched in 2018. Kepler followed an Earth trailing heliocentric orbit. It houses a single differential photometer that with a wide field of view (115 square degrees) that continuously and simultaneously observes the brightness of nearly 150,000 stars. The cadence of these observations are either 1 minute or 30 minutes. The 1 minute cadence data is only downloadable for 512 stars at a time. This is due to the limit on data downloadable from the satellite. This is a problem for asteroseismological study of main sequence stars but is not a major issue for Red Giants as their periods are higher and can be studied well even by the larger cadence data. This allows asteroseismological studies on nearly 16,000 Red Giants from Kepler's Data.

Kepler had two phases of observation. The first phase lasted about 4 years which included measurements of many stars with a suitable duration to perform seismological studies. However, following the failure of two reaction wheels it became impossible to keep the satellite pointed at the same field for long periods. This phase was called K2 and it was still possible to obtain light curves of durations lesser than 5 months but this is not always enough for seismological studies. The data in this study are mostly from the first phase of the satellite's operation.

The data usually consists of a light curve with a duration of 3 to 4 years. This data were downloaded and stored in various repositories. They can be downloaded conveniently using the `lightkurve` python library. This allows us to easily download the time series data from the Kepler mission and perform simple processing to it. As part of the data collection, `lightkurve` is used to download all the data on a particular star, stitch them together, remove outliers and generate a periodogram. This is then used to extract a power spectrum for frequencies. This power spectrum is then fit using MCMC fitting. This gives us frequencies and an uncertainty estimate from the fitting process.

2. Surface Correction

2.1 Introduction

While the CM and CGM models do improve the accuracy of the computed models by a significant amount, they are still 1-dimensional approximations of an intrinsically 3-dimensional process. This leads to a difference in results that cannot be fixed by changing the 1-dimensional approximation used. This shows up as a systematic difference between observed and model frequencies. This can easily be seen in the case of the Sun as there is abundant information on the Sun's oscillatory modes.

Surface correction methods deal with this systematic discrepancy by fitting the difference between the observed frequencies and their corresponding frequencies in the computed model. This process requires a model that is already close to the observed values. This difference is expected to follow a good trend if it is mainly due to surface corrections.

This difference between the model and observed frequencies is usually modelled in two different ways. One is the power law method proposed by Kjeldsen et al. (2008) or the newer method proposed by Ball and Gizon (2014). Both methods work in similar ways but differ in the function used to fit the difference.

In the power law method the function used to fit the difference is

$$\nu_{obs}\left(n\right) - \nu_{model}\left(n\right) = \left(a\left[\frac{\nu_{obs}\left(n\right)}{\nu_0}\right]^b\right) / \ I, \tag{2.1}$$

where ν_0 is a suitably chosen reference parameter and a and b are parameters to be determined. The newer method proposed by Ball and Gizon (2014) uses a different function,

$$\nu_{obs}\left(n\right) - \nu_{model}\left(n\right) = \left(a_{-1}\left[\frac{\nu_{obs}\left(n\right)}{\nu_0}\right]^{-1} + a_3\left[\frac{\nu_{obs}\left(n\right)}{\nu_0}\right]^3\right) / \ I, \tag{2.2}$$

where a_{-1} and a_3 are the parameters to be determined. In both models I is the normalized mode inertia[1] that is defined by Aerts et al. (2010) Eq. (3.140) which is

$$I = 4\pi\frac{\int_0^R\left[\left|\xi_r(r)\right|^2 + l(l+1)\left|\xi_h(r)\right|^2\right]\rho r^2\,dr}{M\left[\left|\xi_r(r)\right|^2 + l(l+1)\left|\xi_h(r)\right|^2\right]}. \tag{2.3}$$

From the GYRE output we use the value $\mathrm{E_{norm}}$. This value after it is suitably normalized is near unity for p modes and large for g modes and somewhere in between for mixed modes. Since all the frequencies we observe in the Sun are p modes this mode inertia will be unity and can be ignored but in Red Giants with observed mixed modes this value becomes important.

[1]It is to be noted that the symbols used for this mode inertia vary and it is also called Q_{nl} in some literature such as Christensen-Dalsgaard's Lecture Notes.

3. MESA Modelling

3.1 Introduction

MESA or Modules for Experiments in Stellar Astrophysics (Paxton et al., 2011) is an open source software suite designed to run experiments in one dimensional stellar evolution. Its modular nature allows users to add any features that are not present through user written hooks and scripts. MESA is a fully featured stellar structure and evolution library that uses various physics and numerical modules. It provides a clean-sheet implementation of Henyey et al. (1959) with automatic mesh refinement, analytic Jacobians, and coupled solution of structure and compositional equations.

MESA first reads the input files and initializes the physics modules to create a nuclear reaction network and access the EOS and opacity data. MESA receives its basic input from two Fortran namelist files. The first file specifies the type of evolutionary calculation to be performed, the type of input model to use, the source of EOS and opacity data, the chemical composition and nuclear network, and other properties of the input model. The second file specifies the controls and options to be applied during the evolution. After loading the specified starting model into memory the evolution loop is entered. In every iteration of this loop is a single timestep that has four basic elements. First, it prepares to take a timestep by remeshing the model if necessary. Second, it adjusts the model to reflect mass loss by winds or mass gain from accretion. This step also adjusts abundances for element diffusion, determines the convective diffusion coefficients and solves for the new structure and composition using a solver. Thirdly, the next timestep is estimated. Lastly in the fourth part the output files are generated. Further details of this process can be found in the Instrument paper Paxton et al. (2011).

MESA includes some saved models for convenience. We will be using the pre computed models for the Sun when performing calibration to both save on time and make the process itself more consistent as the pre main sequence process is a difficult phase to simulate. In the case of stellar grid modelling we will include the pre main sequence in the simulation and create a model with uniform composition with the given mass, a core temperature below $10^6 K$ so no nuclear burning, and uniform contraction for enough luminosity to make it fully convective. Other initial parameters such as initial hydrogen fraction, He fractions, metallicity, etc. are taken from the inlist which is different for each model that is run.

3.2 Canuto-Mazzitelli Model

The most widely used prescription of convection in stellar simulation is the Mixing-Length Theory (henceforth MLT) proposed by Biermann (1948), Vitense (1953) and others. This model works remarkably well for an extremely simple assumption. The model assumes that a fluid packet travels a length given by the mixing length parameter after which it gets mixed. While this wasn't made with any physical system in mind it was shown by Canuto and Mazzitelli (1991) that MLT can be formulated as a special case of the general derivation where the energy spectrum $E(k)$ is taken as a delta function peaked around a wavenumber corresponding to a large eddy. This can be shown to give the same convective flux as the MLT case but with more of a physical reason.

Canuto and Mazzitelli (1991) show that the MLT case is not a bad approximation when dealing with extremely viscous fluids as the energy spectrum naturally approaches a delta function peaked at some wavenumber. This gives some explanation for why MLT is a bad approximation in the stellar case as the stellar case is nearly inviscid.

In the case of the Canuto-Mazzitelli Model (henceforth CM) the energy spectrum is more general and hence is a better approximation of the stellar case. The exact expressions and equations involved in this model and the subsequent model is given in the next chapter where we discuss the implementation of the model in MESA.

3.3 Canuto-Goldman-Mazzitelli Model

The Canuto-Goldman-Mazzitelli Model (henceforth CGM) is an improvement of the CM model which includes the contribution of turbulence itself in the rate of energy input. This makes the model self-consistent and is expected to perform better than the CM model. However, since a self consistent model is a very complex problem implementing it comes with the trade-of of being forced to use simplified versions of the non linear interactions between eddies.

4. Implementing Canuto-Mazzitelli Model in MESA

The CM and CGM models are not implemented in MESA by default. There is a reference implementation by Dr. Warrick Ball, however, this implementation uses the `run_star_extras.f90` file to add in this functionality. This method was not possible at the time as there was a bug in the pre-main sequence part of MESA that prevented implementation of custom convection options. As a result the CM and CGM models were directly implemented into MESA.

4.1 Theory and Implementation

In this implementation we treat mixing length as a fraction α of the pressure scale height i.e.

$$\Lambda = \alpha \cdot H_p. \tag{4.1}$$

While Canuto and Mazzitelli (1991) also suggests the use of z i.e. the distance to the top of the convection zone, this has proven to be difficult to implement and an extra tunable parameter allows us to obtain better fits, while it is noted that this might not always be a good thing. The addition of an option to use z as the mixing length is something that is being worked on.

We start by listing the relevant equations and expressions that are required for the implementation of the CM and CGM models.

$$a_0\Gamma^3 + \Gamma^2 + \Gamma - \delta = 0, \tag{4.2}$$

$$\Sigma = 4A^2\left(\nabla - \nabla_{ad}\right). \tag{4.3}$$

$$a_0\Omega\left(\Gamma\right)\Gamma^3 + \Gamma^2 + \Gamma - \delta = 0, \tag{4.4}$$

where

$$\Omega\left(\Sigma\right) = \frac{\phi}{\phi^{\mathrm{MLT}}}, \tag{4.5}$$

$$\Sigma = 4\Gamma\left(\Gamma + 1\right), \tag{4.6}$$

$$\phi^{\mathrm{MLT}} = \frac{1}{2}a_0\Sigma^{-1}\left[\left(1+\Sigma\right)^{1/2} - 1\right]^3, \tag{4.7}$$

$$\phi = a_1\Sigma^m\left[\left(1 + a_2\Sigma\right)^n - 1\right]^p, \tag{4.8}$$

$$\delta = A^2 \left(\nabla_r - \nabla_{ad}\right), \tag{4.9}$$

$$A = \frac{\Lambda^2}{9\chi}\left(\frac{g}{2H_p}\right)^{1/2}, \tag{4.10}$$

$$a_1 = 24.868, \quad a_2 = 9.7666 \times 10^{-2},$$
$$m = 0.14972, \quad n = 0.18931, \quad p = 1.8503,$$

where Λ is the mixing length, $\chi = K/c_p\rho$ is the thermometric conductivity, g is gravitational acceleration and H_p is the pressure scale height. In this model Λ can be taken as αH_p or z where z is the distance to the top of the convective layer.

The CM model also provides different expressions for convective velocity and turbulent pressure. Equation (41) of Canuto and Mazzitelli (1991) gives the convective velocity as

$$v_t = \left(\frac{\chi}{\Lambda}\right)\sqrt{2S}, \tag{4.11}$$

where S is given by Equation (7) of Canuto and Mazzitelli (1991) as

$$S = (81/2)\,\Sigma. \tag{4.12}$$

The Canuto-Goldman-Mazzitelli Model (Canuto et al. (1996)) is an improvement of the original CM model that uses a self-consistent model for the rate of energy input that takes into consideration the turbulence generated. This manifests as a change in the value of Ω to

$$\Omega_{\mathrm{CGM}} = \frac{\Phi_{\mathrm{CGM}}}{\Phi_{\mathrm{MLT}}}, \tag{4.13}$$

where

$$\phi_{\mathrm{CGM}} = F_1(\Sigma) \cdot F_2(\Sigma), \tag{4.14}$$

$$F_1(\Sigma) = \left(\frac{K_0}{1.5}\right)^3 a\Sigma^k \left[(1 + b\Sigma)^m - 1\right]^n, \tag{4.15}$$

where K_0 is the Kolmogorov constant which we set to 1.5[1] and

$$a = 10.8654, \quad b = 0.00489073, \quad k = 0.149888,$$
$$m = 0.189238, \quad n = 0.185011,$$

and

$$F_2(\Sigma) = 1 + \frac{c\Sigma^p}{1 + d\Sigma^q} + \frac{e\Sigma^r}{1 + f\Sigma^t}, \tag{4.16}$$

where

$$c = 1.080\,71 \times 10^{-2}, \quad d = 3.012\,08 \times 10^{-3}, \quad e = 3.344\,41 \times 10^{-4},$$
$$f = 1.25 \times 10^{-4}, \quad p = 0.72, \quad q = 0.92, \quad r = 1.2, \quad t = 1.5.$$

Equation (4.4) is effectively similar to a cubic equation for both CM and CGM models. However, it is not the same and hence needs to be solved by an appropriate root solving routine. It can be shown that the function is monotonically increasing. This is important as when we do bisection to find the

[1] This value should be a parameter in principle and will eventually be added as an inlist setting.

root we can decide on a bracket for the root by simply finding a point where the function is negative and positive.

However, there are some complications with this function. The δ in the equation varies in magnitude from extremely small ($\approx 10^{-5}$) to very large ($> 10^{12}$). When the value is extremely large the function is sharply vertical around the root which makes determining the exact root in limited precision very difficult.

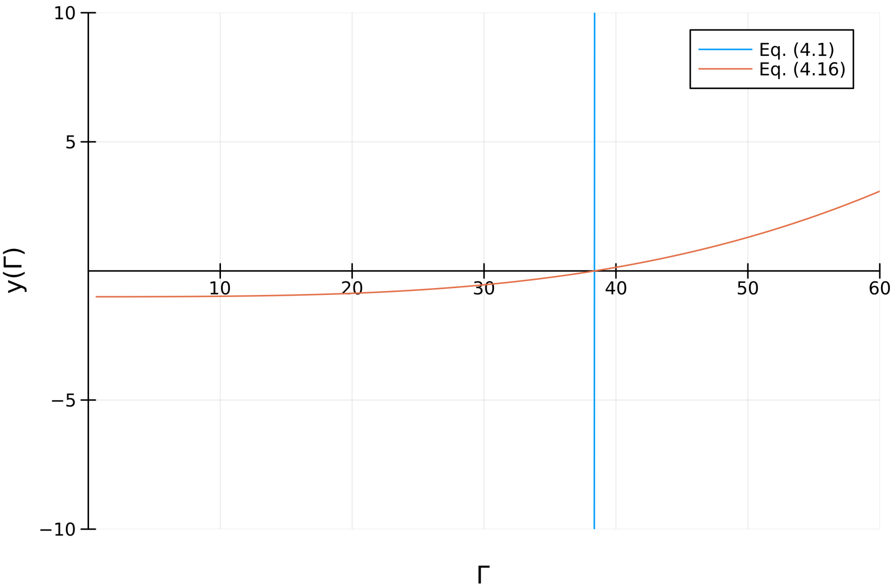


Figure 4.1

To solve this problem we divide the whole equation by δ to normalize the equation. We only do this when $\delta > 1$ to ensure we work with numbers near unity for the whole domain of values. This gives us the following equation that we solve instead:

$$\boxed{\frac{1}{\delta}\left(a_0\Omega\left(\Gamma\right)\Gamma^3 + \Gamma^2 + \Gamma\right) - 1 = 0.} \tag{4.17}$$

We solve the equation to obtain the root using bisection method. This is mostly due to the fact that it is robust and absolutely converges. We would, however, need to compute the derivatives by hand since we aren't using a method that takes advantage of MESA's auto-differentiation but this is not a problem as Dr. Warrick's original implementation includes the required expressions since it was written before the `autodiff` module was written.

For bisection we would normally need to scan the range for a bracket to begin the bisection. However, since we know that the function is monotonically increasing, it is sufficient to find a point where the function is negative and one where it is positive. The lower bound is chosen near zero as the root is usually not much greater than 0. There are a few choices for the upper bound. They are

$$x = \frac{\left(\sqrt{1+4\delta} - 1\right)}{2}, \tag{4.18}$$

and

$$x = \delta^{1/3}. \tag{4.19}$$

The first expression is the root of the quadratic part of the equation which leaves behind a term which is always positive. However, this is not a good upper bound as for larger values of δ this is much larger than the actual root and more iterations are required to converge on the root.

The second upper bound is much better as it is much closer to the root and is guaranteed to be a upper bound. This can be seen if we look at the normalized version of the equation i.e. Eq. (4.17). We see that since each term is divided by δ, if we take $\Gamma = \sqrt[3]{\delta}$ then we can see that the cubic term simply becomes $a_0\Omega(\Gamma)$, this added with the square and linear terms is larger than 1 and hence is always positive, making this a valid upper bound.

As of writing only the first upper bound i.e. Eq. (4.18) is implemented in my branch of MESA.

4.2 Comparison

To test the performance of the two new models, we will calibrate a solar model using the CM and CGM models. We will use these models to calculate the frequencies and compare them to the observed frequencies.

To do this we will use the test suite example of Simplex Solar Calibration that is included with MESA. This uses a simplex search and minimizes a χ^2 that depends on the given parameters of the Sun. We will only consider the log luminosity and log radius as this is enough to ensure good conformance of effective temperature and some other parameters. We set it to use a pre computed late pre-ZAMS model and allow it to converge up to $\chi^2 \approx 10^{-12}$.

Model	Initial Y	Initial FeH	Alpha
MLT	0.2691281717	0.0584289578	1.9119207919
CM	0.2696611716	0.0608622713	0.9824566765
CGM	0.2703242755	0.0640423488	0.8985029878

We see that that the calibrated alpha values are in accordance with what we expect. We expected a lower alpha value than MLT as CM and CGM models provide a larger flux for the same alpha.

These models are then used to generate oscillation frequencies from $l = 0$ to $l = 300$ using GYRE. These frequencies are then matched with observed frequencies computed from HMI's 5 year data. After which we compute the difference between the observed frequencies and the model frequencies for each of the three models. The differences are scaled using Q_{nl} to collapse the multiple curves onto a single curve.

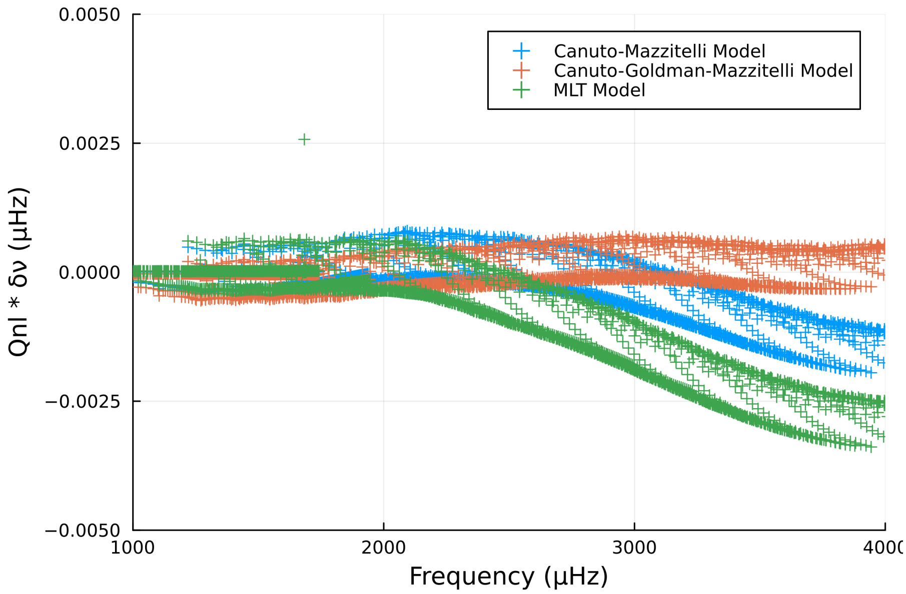


Figure 4.2: Difference in frequencies between observed and model frequencies scaled by Q_{nl}

Here we can see that both the CM and CGM model are closer to the observed frequencies than the MLT model. We also note that at lower frequencies they are all similarly accurate. The major difference between the models is at the higher end of the frequencies. We also note that the CGM model appears to be the best at modelling solar frequencies.

5. Results

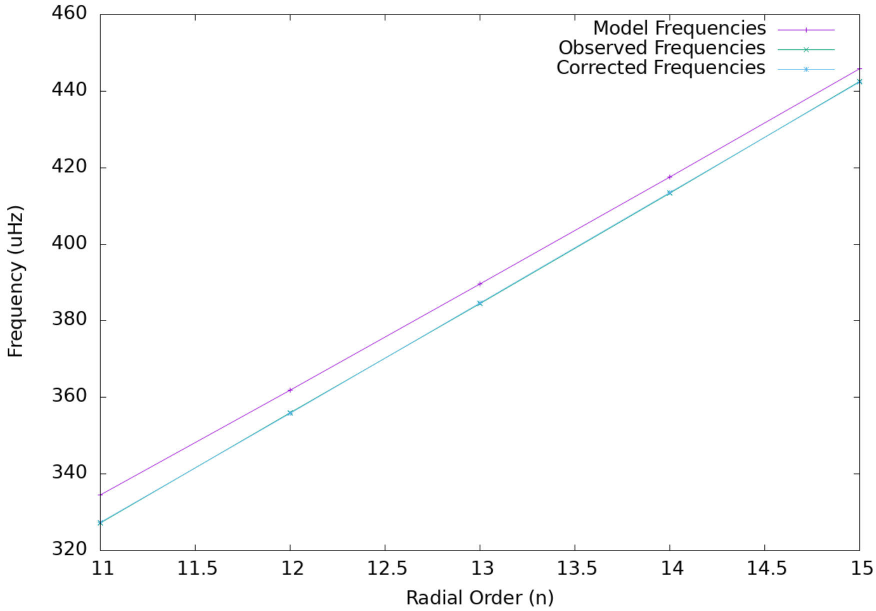
I had run a grid of stellar models as part of my master's thesis. This grid spans the following parameters.

Mass ($M_\odot$)	0.76 – 1.01 with a step size of 0.01
Z_{in}	0.0002, 0.0004, 0.0008, and 0.0010 – 0.0030 with a step size of 0.0005
Y_{in}	0.240 – 0.260 with a step size of 0.001
α_{CGM}	0.6 – 1.4 with a step size of 0.1

All of the models were run using CGM model which was implemented earlier. Using a suitable conditions we obtain the best fit model for the star we are studying (KIC 7341231). We show the effects of surface correction on this model. This model has the following parameters

$$M = 0.78\,\mathrm{M}_\odot, \quad \text{Initial He} = 0.256$$
$$\text{Initial Metallicity} = 0.0004, \quad \alpha = 1.$$

If we apply surface correction (Ball and Gizon (2014) method) to this model's $l = 0$ frequencies we get the following effect



We see that the corrected frequencies match the observed frequencies upto $\pm 0.2\,\mu$Hz. This process was done for all the models prior to calculation of the best fit model.

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